DAY 45 PROJECT REPORT

Sales & Service Demand Forecasting System

Veda Technology — Data Science Internship

Purpose: Build a practical forecasting workflow that analyzes historical demand, identifies trends and seasonal patterns, compares forecast models, and supports service-capacity planning.

Data notice: This project uses synthetic educational data only. It does not contain real Veda Technology records and its forecast must not be presented as a real company prediction.

1. Problem statement

Organizations need to estimate future demand for technology services, digital services, training programs, and other offerings. Forecasts can help plan staff, service capacity, resources, and training seats.

2. Objectives

Preprocess historical time-series data; detect trend and seasonality; visualize daily and monthly demand; build simple baselines; compare statistical and machine-learning approaches; evaluate model accuracy; generate a 30-day forecast; and translate results into business recommendations.

3. Tools and libraries

Python, Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, statsmodels, Jupyter Notebook, and optional Streamlit.

4. Dataset design

The included synthetic dataset contains 1,096 daily records from 2023-01-01 through 2025-12-31. Fields include date, demand, simulated promotion flag, weekday, month, and year. Demand includes simulated trend, weekly variation, annual seasonality, promotion effects, and random noise.

5. Methodology

The notebook validates date order and missing values, plots demand and rolling averages, examines demand by weekday and month, then reserves the latest 90 days as a chronological test window. It compares a naïve baseline, 7-day moving-average baseline, Holt-Winters additive trend/weekly seasonality, and a Random Forest model using lagged demand and calendar features.

6. Evaluation

MAE is the mean absolute difference between actual and predicted demand. RMSE gives larger errors more weight. WAPE expresses total absolute error as a percentage of total actual demand. Lower values are generally better; model choice should also reflect operational costs and stability.

7. Forecast output

The notebook refits Holt-Winters on the available history and creates an illustrative 30-day daily forecast. It saves next_30_day_forecast.csv and model_evaluation_results.csv. The charts and metric table are generated when the notebook is run.

8. Business recommendations

Use forecasts as one input to staff scheduling, trainer allocation, support coverage, resource and license planning, and learning-seat allocation. Compare actual demand with forecasts each week, monitor error metrics, and refresh the model when new observations arrive. Treat promotions as a possible driver, but do not claim causation from synthetic data.

9. Limitations and responsible use

Synthetic patterns are cleaner than many real business processes. Real demand may shift because of pricing, competitor actions, launches, policy changes, holidays, and one-off events. No prediction intervals or external business drivers are included. The Random Forest holdout evaluation uses lag features built from observed demand; a production multi-step forecast needs a carefully designed recursive or direct-horizon evaluation. Do not present this educational output as a real company forecast.

10. How to run

Install the dependencies listed in README.md, place the CSV beside the notebook, open the notebook in Jupyter, and run all cells from top to bottom. For the optional dashboard, copy the final Streamlit code cell into app.py and run streamlit run app.py.

Project deliverables

- Jupyter notebook with preprocessing, EDA, baseline and model comparison
- Synthetic historical demand CSV
- Forecast and evaluation CSV outputs generated by the notebook
- README with setup and run instructions
- Optional Streamlit dashboard starter

Prepared for Veda Technology — Data Science Internship, Day 45.